

Contemporary Misdiagnosis of an Emergent Enemy Military "Stack"

A strict-criterion historical comparison for *The Artillery State*, on the question of whether English observers of the loss of Normandy (1449–50) failed to perceive an integrated French fiscal-standing-army-artillery-siege-engineering-capitulation system.

Compiled 9 August 2026. Every factual claim below is grounded in a page fetched during this research session, cited inline with the exact URL.

1. The inclusion test applied

A case is admitted only if all six conditions hold:

1. **Contemporaneity.** A defeated-side account written or spoken during the conflict or within roughly ten years.
2. **Explicit causal claim.** The account offers an identifiable explanation of why defeat happened.
3. **Demonstrated error.** Modern scholarship shows that explanation was materially incomplete or wrong.
4. **Integrated novel stack.** The missed cause is a *combination* — organisation + doctrine + logistics/finance/communications + weapons — not a single superior weapon or generic numerical inferiority.
5. **Perception without comprehension.** Evidence that the observers either described individual components while missing their combined effect, or treated enemy output as inexplicable and filled the gap with providence, cowardice, treason, luck, numbers or morality.
6. **Not knowing propaganda, not retrospective legend, not a correct inquiry.**

Condition 6 turns out to be the sharpest blade. It removes most of the famous "stab-in-the-back" cases, and — importantly for the argument — it also removes several cases that intuition suggests should qualify (Prussia 1806, France 1870, Russia 1905), because in each of those the defeated state's own inquiry *did* correctly name the enemy system within a few years. That is exactly what makes the surviving cases analytically valuable: they are the ones where the defeated side had the components in front of its eyes and still could not assemble them into a system.

2. Ranked final set

Rank 1 — Mamluk Sultanate vs. Ottomans, Marj Dabiq and Raydaniyya, 1516–17

Defeated-side account: Muhammad ibn Ahmad Ibn Iyas, *Bada'i' al-zuhur*, eyewitness Cairo chronicle written as events occurred; English translation W. H. Salmon, *An Account of the Ottoman Conquest of Egypt in the Year A.H. 922 (A.D. 1516)*, Oriental Translation Fund, 1921 — [full text at archive.org](#).

What Ibn Iyas demonstrably knew. He is precise about the Ottoman components. At Marj Dabiq the Mamluk amir Karanisah captured "the guns on the carriages, and the musketeers" (p. 42). Before Raydaniyya, Sultan Tumanbay assembled "about thirty or more" ox-drawn wagons carrying musketeers, camel-mounted musketeers and wooden shields (p. 81), and displayed "great energy in making wagons and forging guns, in the manufacture of muskets, and the raising of a very large number of musketeers" (p. 92), fortifying Raydaniyya with cannon, shields and a trench (p. 98). This is the decisive evidentiary feature of the case: **the Mamluks identified the enemy's weapons, procured them, replicated the wagon-fort, and were still annihilated.**

The causal explanation he gave. Three registers, none of them systemic:

- *Divine decree:* "this loss was inflicted on the Egyptian troops by the will of Providence in fulfilment of His decrees" (p. 42); "God's anger was kindled against the Egyptian troops, and their hands seemed to be fettered" (p. 43); "Thus was it written in the Divine decrees" (p. 46); after all Tumanbay's preparations, "But Allah did not grant him the victory" (p. 97).
- *Moral desert:* "Neither the Sultan nor his Amirs ruled the people justly; they reaped the reward of their evil doings" (p. 46).
- *Treason and factional failure:* Khair Bey was "secretly in league with Ibn 'Othman" and "first to fly before all the troops," whom Selim "called ... Khain Bey (traitor)" (pp. 42, 51), explicitly assimilated to Ibn al-'Alkami's alleged betrayal at Baghdad; and the imported mamluks ordered not to fight "might have been blocks of wood for all the stir they made" (p. 44). Where a material cause is offered it is bare number: "Courage cannot avail against numbers" (p. 66).

The stack modern scholarship identifies. Gábor Ágoston, "Firearms and Military Adaptation: The Ottomans and the European Military Revolution, 1450–1800," *Journal of World History* 25, no. 1 (2014): 85–124, argues that "in addition to firearms, factors such as good intelligence; resourceful leadership; large and disciplined professional armies; superior supply and logistics; and the combined use of field artillery, infantry firepower, wagon fortresses, and cavalry charges were all important in Ottoman military successes" (p. 87), that "their main advantage lay in their early integration of gunpowder weapons into their standing forces" (p. 93), and that behind it stood "the strength of the Ottoman administrative-fiscal and logistical systems" (p. 111) — "administrative, fiscal, and logistical capabilities to match their military might" (p. 123) ([Ágoston, JWH 25.1, PDF](#)). On Raydaniyya specifically, Ágoston notes that "the Ottomans learned about Tumanbay's plans through their spies and captured Mamluk soldiers and altered their tactics accordingly" — i.e. the intelligence layer defeated the copied hardware layer. Albrecht Fuess, "Les janissaires, les mamelouks et les armes à feu," *Turcica* 41 (2009): 209–

227, independently dismantles the old story that the Mamluks lost through a chivalric refusal of firearms ([Peeters Online Journals](#)) — which is precisely why the "they wouldn't use guns" explanation cannot be substituted for the systemic one.

Why the contemporary explanation was wrong. Ibn Iyas's causes (God, injustice, one traitor, factional indiscipline) are each locally true but jointly cannot explain why the *same* weapons in Mamluk hands produced nothing. The missing variable is not the musket but the standing, salaried, drilled infantry corps that could deliver sustained volleys in formation, financed and supplied by a fiscal-administrative state and directed by an intelligence apparatus. Ibn Iyas has no vocabulary for a permanent professional infantry establishment as a *system property*; he treats troops as a quantity of bodies and courage.

Classification: genuine conceptual blindness, expressed through an ideological-theological framing (divine decree) with a secondary scapegoating overlay (Khair Bey). Not propaganda: the chronicle is private, bitter, and freely critical of the Mamluk sultans themselves.

Confidence: High. **Analogy to Normandy 1449–50:** Very close. Both are cases of a militarily competent élite that *acquired the enemy's hardware* (Mamluk wagon-guns; English attempts to ship artillery and powder into Normandy) and still lost, because the hardware was the visible tip of a fiscal-organisational system. Curry's finding that Somerset told parliament in 1449 that "there is no place in the kings obeissaunce there purveied nother in reparations, ordenaunce ne in eny maner artillerie," that remedying it "wold drawe to inestimable costs," and that "even where there were guns there was a shortage of powder" ([Curry, "Guns and Goddams," p. 185](#)) is the same structure of complaint: the object is named, the system behind the object is not.

Rank 2 — France after Crécy and Poitiers, 1346–57

Defeated-side accounts (all within 1–11 years of the events):

- Jean de Venette, *Chronicle* (contemporary French Carmelite, writing through the 1340s–60s) — [text](#).
- Count of Armagnac's letter of 1 October 1356; the anonymous *Complainte sur la bataille de Poitiers* (1357, ed. Charles de Beaurepaire, *Bibliothèque de l'École des chartes* XII (1851): 257–63); François de Montebelluna, *Tragicum argumentum de miserabili statu regni Francie* (1357); Guillaume de Machaut, *Le confort d'ami* (1357) — all analysed with extensive quotation in Sylvie Lefèvre, "La déconfiture," [Presses universitaires du Septentrion / IRHiS, OpenEdition](#). The Beaurepaire edition of the *Complainte* is catalogued at [Persée](#).

What the accounts demonstrably knew. Venette knows the English fought with massed archery and that they protected their bowstrings in the rain while the Genoese crossbows were ruined: the English "had quickly protected their bows by putting the bow strings on their heads under their helmets" (p. 43). He knows the French attacked "in some disorder and confusion, on account, it is said, of undue haste on the part of the French king." At Amiens he records that "the English, being much more numerous and defending themselves with arrows, prevailed" (p. 42). The *Complainte* knows that the French pay system was being defrauded: "Leurs soillars et leurs pages pour gens d'armes contoient / Ainsi un seul pour quatre gages du roy prenoient,"

and — remarkably — urges the king to enlist "Jaque Bonhomme," i.e. paid commoners, plainly modelled on the enemy's archers.

The causal explanations given. (a) *Treachery misread from a technical fact*: seeing the Genoese fail, "Our Frenchmen, seeing this and not knowing the cause or impediment, supposed that they were feigning treacherously... They accordingly began to kill and massacre them" (Venette, p. 43) — a documented case of a mechanical cause being converted directly into a treason narrative in real time. (b) *Noble cowardice and selfishness*: the nobles fled "either from cowardice or from love of ease," and "despised and hated all others... In no wise did they defend their country from its enemies" (Venette, ed. intro., pp. 4–6); chroniclers hold that "la cause évidente de la défaite... est la lâcheté de ceux qui 's'en fouyrent honteusement et vilainement'" ([Lefèvre](#)). (c) *Providence and sin*: Armagnac, "aychi cum a Dio a plagut a suffrir, lo rey Mossenhor es estat desconfit"; Montebelluna attributes the disaster to "le juste jugement de Dieu" and a catalogue of the sins of clergy, nobility and people — "ce sont nos péchés qui vous ont faits ce que vous êtes." (d) *Fortune*: Machaut's wheel.

The decisive quotation for the thesis. Montebelluna states the anti-systemic position explicitly: "Les vices de l'armée féodale ne viennent pas de sa structure, mais de la lâcheté avant tout" ([Lefèvre](#)). A contemporary considers the structural hypothesis and rejects it in favour of a moral one. That is the purest documented instance in this set of a contemporary refusing a systems explanation.

The stack modern scholarship identifies. Clifford J. Rogers, "The Military Revolutions of the Hundred Years War," *Journal of Military History* 57, no. 2 (1993), describes English armies "drawn from the common population...; they served for pay; they fought primarily on foot, in close-order linear formations which relied more on missile fire than shock action; and they fought to kill," deploying "the 'pike-and-shot' combination of dismounted men-at-arms and archers which they had developed in the 1330s in Scotland," with which "the English at Crecy devastated a French force... approximately three times as numerous"; and he insists that "what made these horsemen on foot so effective was their integration with the archers" ([De Re Militari full text](#)). Andrew Ayton's reconstruction of the Crécy army confirms the mixed retinue-and-array recruitment structure behind it ([Historical Research](#) 92 (2019): 24).

Why the explanation was wrong. Every French cause is a moral or accidental one attached to a French actor. None of them is a description of the enemy. The French correctly registered rain, disorder, haste, arrows and numbers, but the causally operative fact — that a cheaply recruited, wage-paid missile arm was *integrated* with dismounted men-at-arms into a defensive combined-arms system, and that this rested on a contractual military-fiscal organisation — is nowhere named. The *Complainte* comes closest by proposing to hire commoners, but frames it as a remedy for pay fraud and noble dishonour rather than as adoption of an enemy system.

Classification: ideological-theological framing plus genuine conceptual blindness; the *Complainte* shows partial component-level perception. **Confidence:** High. **Analogy to Normandy:** Direct — same war, same politics, roles reversed a century later, and the *Complainte*'s partial insight is the mirror image of Somerset's partial insight in 1449.

Rank 3 — Qing China, First Opium War, 1839–42

Defeated-side accounts: Qing memorials, edicts and officers' reports, quoted from Alaric Searle and Zhang Yi, "Soldiers with stiff bodies': rumors, stereotypes and the Chinese image of the British army during the First Opium War (1839–1842)," *Journal of Modern Chinese History* 14, no. 1 (2020): 86–110 ([Taylor & Francis](#)).

What the accounts knew, and the causal explanation given. Qing observers recognised British naval and artillery superiority and stopped there: "the barbarians' advantages were their cannons and ships and land fighting was not [one of their strengths]," and the Daoguang Emperor held that "the British relied basically on their formidable ships and cannon fire to harass coastal provinces" (p. 90). Beyond that came a physiological fantasy — "the soldiers of the barbarians were not good at fighting because their legs and feet were bound so tightly that they could not move freely," so on land they "would be embarrassed just like fish on solid ground" (p. 89); "their cannons could not work on mountains, their bayonets could not be used to attack at a distance, their bodies were stiff" (p. 91); "once they had fallen, they would never get up again" (p. 90). Searle and Zhang observe that "most Chinese officers were satisfied by this simple explanation of their superiority" (p. 92). Defeats were then attributed to *hanjian* (traitors) and to Indian troops, and reversals were denied outright: "the powerful ships and cannons are all lies and the high mast fake"; "the barbarians are not tigers and wolves, but pigs and deer instead" (pp. 94–97).

The stack. Searle and Zhang document the actual British army as an institution: "famous for its rigid discipline," advancing "with 75 steps to the minute, after repetitious training," and delivering volleys at 100 yards (pp. 102–103) — i.e. a drilled professional infantry, not stiff-legged invalids. That infantry was joined to an industrial naval component whose emblem was the *Nemesis*: built by John Laird at Liverpool and "launched on the Mersey late in 1839... for the Secret Committee of the East India Company," "except for deck, spars, and sundries... built entirely of iron," 184 feet by 29, flat-bottomed with a maximum draught of six feet, twin paddle engines, and armed with "two pivot-mounted thirty-two-pounders, several six-pounders and swivels, and a rocket launcher" — "at the core of this British military superiority lay iron and steam, the key components of that Industrial Revolution" ([Victorian Web](#)). The shallow draught is the systemic point: it converted naval power into river-and-estuary power, which converted into interdiction of the Grand Canal and the tax-grain artery. And the British ran an intelligence layer — purchase of the *Peking Gazette*; Lin Zexu's own translation unit is the mirror-image Chinese attempt (Searle and Zhang, pp. 95, 104).

Why the explanation was wrong. The Qing diagnosis was component-level and static: guns and ships are things the enemy *has*, and their supposed inability to fight on land is a property of their bodies. Missing is the integration — iron steamers with shallow draught projecting drilled infantry and artillery up rivers, sustained by industrial production, imperial logistics from India, and systematic intelligence. The post-war Chinese response confirms the diagnosis stopped at technology: Wei Yuan's *Haiguo tuzhi*, first printed in 1842 and expanded to 100 *juan* by 1852, concluded that "China was to learn from the advanced technology of the barbarians in order to check them" and that "foreign naval, military and industrial technology was to be adopted by China," with armies modernised through "better training and professionalization" ([ChinaKnowledge](#)). This is the strongest form of the case: even the *best* contemporary Chinese

analyst framed the lesson as technology transfer plus training, not as an integrated industrial-fiscal-institutional system — which is why *zhiqiang* self-strengthening bought ships and lost again in 1894–95.

Classification: genuine conceptual blindness at élite level, with an explicit scapegoating layer (*hanjian*) and a folk-stereotype layer (stiff legs). Some memorial-writing is doubtless careerist self-protection, which is why this ranks third rather than first. **Confidence:** High for the content of the beliefs; Medium on sincerity for individual memorialists. **Analogy to Normandy:** Strong on the "we saw the guns" axis; weaker in that the technological gap in 1841 was genuinely categorical, whereas in 1449–50 both sides had guns.

Rank 4 — Kingdom of Hungary vs. the Mongols, 1241–42

Defeated-side accounts: Master Roger of Torre Maggiore, *Epistle to the sorrowful lament upon the destruction of the kingdom of Hungary by the Tatars*, written by a survivor and Mongol captive c. 1243–44; and Thomas of Spalato, *Historia Salonitanorum*, begun c. 1245–51. Bilingual CEU Press edition of Master Roger: *Anonymus and Master Roger*. Analysis of Thomas: *Florilegium 4* (University of Toronto Press).

What the accounts demonstrably knew — an extraordinary inventory. Master Roger describes: a deliberate four-to-five-day Mongol withdrawal to preserve fodder and prevent news reaching Hungary (p. 165); a multi-axis converging advance by Batu, Qadan and Peta (pp. 165–67); Qadan's three-day forced march through forest (p. 167); repeated feigned retreat (pp. 167, 169, 179, 181); the deliberate opening of an escape corridor at Mohi — "The Tatars stepped back a little and opened up among themselves a point of exit where they did not shoot" (p. 183); scouts (pp. 203, 225); deception with unguarded horse herds to test whether the Danube ice would bear weight (p. 215); an occupation administration of *kneses* or "bailiffs" each set over roughly a thousand villages (p. 209); seven siege engines bombarding day and night (p. 201); thirty siege engines at Esztergom behind a wall of twig bundles built by prisoners (p. 217). Thomas of Spalato adds armour, arrows, small hardy horses "satisfied with a little straw for fodder," coracles, night attack, encirclement, the Pest reconnaissance and the frozen-Danube crossing, and concedes that "there are no people in the world that have such skill in warfare" (*Florilegium 4*).

The causal explanations given. Providence and sin: Master Roger opens by inviting his readers to "give thanks to the King of Kings, who, forgetful of His mercy, did not spare His oppressed people... He visited their iniquities not with a rod but with stripes" (pp. 132–35). Thomas calls the Mongols a "pestilential nation," "not men, but devils," and concludes: "And by divine judgment the Hungarians suffered inordinately the punishment of their sins" (*Florilegium 4*). Then domestic political causes: nobles "would have liked the king to be defeated so that they would then be dearer to him" (ch. 28, p. 181); Hungarians "trusting their number, made fun of all this and... had neither mind nor spirit for fighting" (p. 181); Cuman–Hungarian enmity (ch. 3, p. 141); a "fainthearted" king (p. 169); refusal to believe warnings (ch. 14, p. 157); the erosion of county resources (ch. 10, pp. 151–53); and the betrayal by the duke of Austria (ch. 32, pp. 192–95).

The stack. Modern scholarship treats the Mongol force as an integrated operational system rather than a raid: they "employed hunting tactics in war... they advanced against Europe in *jerge* formation, a hunting circle or battue," a term for "an encircling movement" used with reference to "the military application of the *nerge* formation and the steppe tradition of hunting as military training"; terror was instrumental — "the Mongols may well have employed unprecedented cruelty as a tactic to demoralize their enemies and stifle resistance," with prisoners used as "arrow fodder"; and famine was a weapon — "the famines the Mongols triggered were intentional—a sort of weapon to crush resistance," "basically a man-made famine" (József Laszlovszky, Stephen Pow, Beatrix F. Romhányi, László Ferenczi and Zsolt Pinke, "Contextualizing the Mongol Invasion of Hungary in 1241–42," *Hungarian Historical Review* 7, no. 3 (2018): 419–450, at pp. 423, 428, 433–34, 442 — [PDF](#)). The same authors state the perceptual point directly: the Hungarians "encountered a little-known and poorly understood enemy—not a raiding band of steppe horsemen, but a well-organized and large army attacking the country with the intention to subjugate or destroy the population" (p. 442).

Why the explanation was wrong. Master Roger's inventory of components is almost complete; his causal account contains none of them. Feigned retreat, multi-axis convergence, remount economy, reconnaissance, siege train with prisoner labour, and immediate occupation administration appear in his narrative as episodes and atrocities, never as an explanation. The explanation slot is filled by God, royal weakness, noble disloyalty and Cuman hatred.

Caveats that keep this at rank 4. First, the Laszlovszky team explicitly argues against a monocausal Mongol-superiority story: "any monocausal explanation appears to be insufficient to explain the withdrawal," there was "more resistance to the invasion than has hitherto been acknowledged," Mongol objectives "were being foiled," and Mongol-derived material in the *Yuan Shi* records that Mongol princes "wished to flee from the country already during the Battle of Muhi" (pp. 437–38, 442). So the defeated-side accounts are wrong about *causation*, but the Mongol system was not omnipotent. Second, the domestic political causes Master Roger names (Cuman expulsion, magnate obstruction) are partially true. The case therefore satisfies "materially incomplete" rather than "flatly wrong."

Classification: ideological-theological framing plus genuine conceptual blindness; the vocabulary for a permanent, staff-directed, logistically planned army did not exist in Latin Christendom in 1243. **Confidence:** High on the texts, Medium-High on the analytic claim. **Analogy to Normandy:** Moderate — the Mongol stack was far more alien than the French one, so the failure to conceptualise it is less surprising and therefore less probative for the Normandy argument.

Rank 5 — France, May–June 1940

Defeated-side account: Marshal Philippe Pétain, radio address of 20 June 1940. Full text: lignemagnot.com. Pétain frames the speech as an answer to the causal question — "Cet échec vous a surpris. Vous souvenant de 1914 et de 1918, vous en cherchez les raisons. Je vais vous les dire" — and then gives an entirely quantitative and demographic answer: on 1 May 1917 "nous avions encore 3 280 000 hommes aux armées," whereas "à la veille de la bataille actuelle, nous en avions 500 000 de moins"; "en mai 1918, nous avions 85 divisions britanniques: en mai 1940, il n'y en avait que 10," plus in 1918 "les 58 divisions italiennes et les 42 divisions américaines";

"l'infériorité de notre matériel a été plus grande encore que celles de nos effectifs"; "l'aviation française a livré à un contre six ses combats." The conclusion is the formula: "**Trop peu d'enfants, trop peu d'armes, trop peu d'alliés : voilà notre défaite.**" A moral supplement follows: "Depuis la victoire, l'esprit de jouissance l'a emporté sur l'esprit de sacrifice." Two days later at Bordeaux the same explanation was given for foreign consumption: France, "inferior in arms and numbers, ... had to ask for the combat to cease" ([ibiblio, Pearl Harbor Attack policy documents](#)).

The internal witness that this was the general diagnosis. Marc Bloch, writing in 1940, records the official causal menu with contempt: "Nous venons de subir une incroyable défaite. A qui la faute ? Au régime parlementaire, à la troupe, aux Anglais, à la cinquième colonne, répondent nos généraux. A tout le monde, en somme, sauf à eux" (*L'étrange défaite*, p. 45, [UQAM Classiques des sciences sociales PDF](#)).

The stack. Bloch himself names it, in systemic terms: "Nos chefs ou ceux qui agissaient en leur nom n'ont pas su penser cette guerre... le triomphe des Allemands fut, essentiellement, une victoire intellectuelle" (p. 56); "Les Allemands ont fait une guerre d'aujourd'hui, sous le signe de la vitesse... nous n'avons pas su ou pas voulu en comprendre le rythme"; "c'est nous, cette fois, qui jouions les primitifs" (pp. 56–57); "le métronome des états-majors ne cessa pas de battre plusieurs mesures en retard" (p. 62); the Germans "conservèrent la fâcheuse habitude d'apparaître là où ils n'auraient pas dû être. Ils ne jouaient pas le jeu" (p. 67); their method was a "méthodique opportunisme" — probe, halt where resistance is firm, and where they struck "dans du mou," charge (p. 68). Karl-Heinz Frieser's reconstruction supplies the mechanism: "the Germans triumphed, not through faster tanks or better aircraft but through superior command and control"; "the German army had done a great deal of hard work in the interwar period on the military use of radios, and had at least 12 times the number of trained radio operators as the French"; the operational corps "tended to lead from the front (*führen nach Vorn*)"; command rested on *Auftragstaktik*, in which "the commander devised a mission (*Auftrag*), explained it in a short, crisp order, and then left the method of achieving it to the officer on the spot"; and the result was that "German orders were typically acted upon within minutes, while it rarely took the French less than two or three days to do anything" ([HistoryNet review of *The Blitzkrieg Legend*](#)). Frieser also insists the campaign plan was "a go-for-broke gamble... the panzer thrust through the Ardennes," so reckless that "even a decent performance by the Allies should have stopped it cold" — which destroys Pétain's material-inferiority thesis at the root.

Why the explanation was wrong. Pétain's variables are stocks (births, guns, allied divisions); the operative variable was a rate (decision cycle time) produced by radio density, doctrine and delegated command. The French had comparable or superior tank numbers in several categories and were beaten by tempo, not by inventory.

Classification: *elite scapegoating with uncertain sincerity, shading into genuine conceptual blindness.* This is why the case ranks fifth rather than higher. Pétain had an obvious political interest in a demographic-decadence narrative that indicted the Third Republic, and Bloch's testimony that the generals blamed "everyone except themselves" points to self-exculpation. Crucially, condition 5's "no adequate vocabulary" defence is weakened by the fact that Bloch, a reserve staff officer with no special access, produced the correct systemic analysis within months: the conceptual vocabulary was available in France in 1940. The case therefore illustrates *motivated* rather than *unavoidable* blindness. It does not fall to the exclusion in condition 6 — unlike 1918

Germany, there is no evidence Pétain privately held the tempo analysis and suppressed it — but it should be marked borderline in the book.

Confidence: High on texts, Medium on classification. **Analogy to Normandy:** Strong on form (input-accounting instead of system-accounting: too few men, too few guns, too little money is almost word-for-word the Lancastrian complaint), weaker on sincerity.

3. Borderline case, included with a warning

Rank 6 (borderline) — Qing China, Sino-Japanese War 1894–95, as the *repeat* of Rank 3

This is included not as an independent case but as the confirmatory second iteration of the Opium War failure. Its evidential value is that the Chinese remedy adopted after 1842 was exactly the component-level one Wei Yuan prescribed — adopt "foreign naval, military and industrial technology," modernise armies with "better training and professionalization" ([ChinaKnowledge](#)) — and that this technology-transfer reading of the first defeat did not prevent the second. I did not fetch contemporary 1894–95 Chinese primary accounts in this session, so no independent causal quotation is asserted here. Treat as a research lead, not as an established case.

4. Rejected candidates and why they fail

Candidate	Fails test	Evidence
Prussia 1806 (Jena–Auerstedt)	3 and 6 — the defeated side diagnosed correctly and immediately	The Prussian reformers "clearly analyzed the military, social, and political weaknesses that had led to the defeat" and reformed "organization, leadership, education, and jurisdiction," pursuing "a new structure of the army... the establishment of a Ministry of War including a modern general staff, or the implementation of the universal conscription"; Scharnhorst had already before 1806 wanted "universal military service through short-term conscription, changes in the strategic and tactical doctrine, and a general staff," promotion by merit, and "the adoption of new formations like divisions and corps" (DTIC, <i>The Reorganization of the Prussian Army After 1807</i>, pp. ii, 1–4). The correct systemic answer existed inside the Prussian army <i>before</i> the defeat; the failure was political, not conceptual.

Candidate	Fails test	Evidence
France 1870–71	6 — the official inquiry explicitly rejected the treason narrative and named the systemic causes; and the reform correctly imitated the enemy system	"À peine les combats terminés, une commission parlementaire présidée par l'amiral Jauréguiberry enquête sur ses causes : elle conclut à la faiblesse des effectifs et à l'insuffisance de l'encadrement de l'armée, mais écarte les rumeurs de 'trahison'." Analysis then moved to "des causes morales et intellectuelles, qui invite à s'intéresser à l'organisation militaire, politique et culturelle du vainqueur," with Prussia becoming "un modèle à méditer." The laws of 27 July 1872 and 24 July 1873 rebuilt recruitment, mobilisation, territorial corps organisation and peace–war continuity, and War Minister General du Barail conceded: "Nous avons plus imité que créé" (Cairn, " La réforme de l'armée française après 1871 "). The popular " <i>nous sommes trahis</i> " cry and the 1873 Bazaine court-martial were real — Varley notes "initially there was a search for scapegoats... In 1873 came the court-martial of the 'traitor' Marshal Bazaine" (<i>French History</i> 24, no. 1: 122) — but the institutional diagnosis was correct within eighteen months. This is the Argentina-1982 pattern.
Russia 1904–05	6 — correct institutional diagnosis within a year	"The war with Japan in 1904–1905 had revealed the navy's many weaknesses—in particular, that it lacked both a plan for fighting a war in the Far East and the facilities needed to support a fleet there." Lieutenant A. N. Shcheglov's 1905 memorandum "The Importance and Work of a Naval General Staff on the Basis of the Experience of the Russo-Japanese War" produced the Naval General Staff on 7 May 1906, tasked with war planning, army–navy coordination, intelligence, basing, force structure and training (<i>International Journal of Naval History</i>). Stoessel and Rozhestvensky were blamed, but the state simultaneously built the correct systemic answer.
Germany 1918	6 (explicit exclusion) — knowing fabrication by leaders who understood the military situation.	Excluded by the brief's own terms; not researched further.
Aztec omens 1519–21; the Kerkoporta postern at Constantinople 1453	1 — the explanatory myths are later constructions, not contemporary causal accounts.	Excluded by the brief's own terms; not researched further.
Argentina 1982	6 — the Rattenbach inquiry identified the integrated failure correctly.	Excluded by the brief's own terms; not researched further.
Arab states 1967	5/6 — cannot be separated from knowing propaganda without evidence I did not obtain in this session.	Not established; leave out.

Candidate	Fails test	Evidence
Carrhae 53 BC; Adrianople AD 378	1 and 4 — no genuinely contemporary defeated-side causal account survives for Carrhae (Plutarch and Dio are far later), and Ammianus, though near-contemporary for Adrianople, gives a broadly accurate account of Roman command error and cavalry surprise rather than a mis-explanation.	Not established from fetched sources in this session; excluded on evidentiary grounds rather than asserted as fact.
Manzikert 1071; Nicomolis 1396	4 — the contemporary explanations (betrayal by Andronikos Doukas; Frankish impetuosity) are substantially <i>correct</i> , so there is no missed stack.	Not researched in depth this session; flagged as unpromising.
Songhay at Tondibi 1591	1 — the <i>Ta'rikh al-Sudan</i> and <i>Ta'rikh al-Fattash</i> postdate the event by roughly two generations.	Chronological fail on inspection; not pursued.

5. Comparative synthesis: what actually blocked perception

Across the five admitted cases the same five mechanisms recur, and they are not the same thing as ignorance.

1. Components are visible; integration is not. In every case the defeated side had an accurate inventory. Master Roger describes feigned retreat, converging columns, remount economy, siege trains and occupation bailiffs. Ibn Iyas describes wagon-guns and musketeers and then *builds thirty of his own wagons*. Venette describes massed archery and the bowstring trick. Qing memorials describe ships and cannon. Pétain counts divisions and aircraft. What none of them describes is a *relation between components*. An integrated system is not an object in the visual field; it is a set of couplings — pay to permanence, permanence to drill, drill to volley, volley to formation, formation to combined arms, and finance underneath all of it. Pre-modern (and, in 1940, hurried) observers had rich vocabularies for objects, persons and virtues, and almost none for couplings.

2. Emergence defeats the additive model of strength. Every contemporary explanatory framework here is additive: $\text{strength} = \text{number of men} \times \text{courage} \div \text{sin}$. "Courage cannot avail against numbers" (Ibn Iyas, p. 66) and "trop peu d'enfants, trop peu d'armes, trop peu d'alliés" (Pétain) are the same sentence 424 years apart. An additive model cannot represent superlinear returns from integration, so when a smaller or equally equipped force wins decisively, the additive model must either invent a hidden subtraction on one's own side (treason, cowardice, sin) or a hidden addition on the enemy's (devils, divine favour, "they don't play the game"). Bloch caught the phenomenon exactly: "les Allemands conservèrent la fâcheuse habitude d'apparaître là où ils n'auraient pas dû être. Ils ne jouaient pas le jeu" ([Bloch, p. 67](#)).

3. Distributed causation has no author, and these genres require an author. Chronicle, lament, sermon, memorial and radio address are all agent-centred genres. A system has no agent to blame or absolve. The genre therefore forces the explanation into the nearest available agent: a traitor (Khair Bey, the *hanjian*, Bazaine, the *cinquième colonne*), a class (the nobles in the *Complainte*), a ruler, or God. This is why the same explanation appears in cultures with nothing else in common — it is a property of the source genre as much as of the culture. The one contemporary who reached the systems answer, Marc Bloch, was writing in a genre he had to invent for the purpose, an unpublishable private *déposition*.

4. Component-level imitation is a trap, not a corrective. The most damaging finding for any "they simply didn't look" hypothesis is that three of the five defeated sides *did* look and *did* copy. Tumanbay built wagon-forts and cast guns and still lost, because the Ottoman advantage lay in "early integration of gunpowder weapons into their standing forces" and in "the strength of the Ottoman administrative-fiscal and logistical systems" (*Ágoston*, pp. 93, 111) — none of which is procurable. Wei Yuan prescribed adopting "foreign naval, military and industrial technology" (*ChinaKnowledge*), and China bought ships. The *Complainte* proposed to hire "Jaques Bonhomme." Copying the visible layer reinforces the belief that the visible layer *was* the cause, and thus insulates the misdiagnosis from disconfirmation.

5. Political incentive selects among available explanations rather than creating them. The rejected cases prove that incentive alone does not determine outcome: France in 1871 and Russia in 1905 both had furious scapegoating politics (Bazaine's court-martial; Stoessel's) *and* produced correct systemic diagnoses within two years, because a professional institution existed whose job was to produce one. Where such an institution did not exist — Mamluk Cairo, Valois France in 1357, Daoguang's memorial system — scapegoating had no competitor. 1940 France sits between: the institution existed and failed, and the correct answer came from outside it.

Did they lack a concept equivalent to systems integration? For 1241, 1357 and 1516 the answer is essentially yes: there is no term in Latin, Occitan or Arabic chronicle vocabulary of the period that does the work of "combined arms," "operational tempo," "military-fiscal state" or "force structure." Roger and Ibn Iyas can name every part and still have no noun for the whole. For 1841 the answer is partly no — the concept existed but only as "technique," which is why the lesson drawn was technology transfer. For 1940 the answer is clearly no: the concept existed, Bloch stated it, and it was rejected for reasons of interest and pride. The historical sequence is therefore not "ignorance gives way to knowledge" but "no concept → the wrong concept (technique) → the right concept, refused."

6. Recommendation for *The Artillery State*

Use these three, in this order.

- 1. The Mamluks and the Ottomans, 1516–17 (Ibn Iyas).** The best case in the set, and structurally the closest to Normandy. A competent military élite watched the enemy's guns, bought and built the same guns, deployed them behind a prepared fortified line, lost anyway, and explained the loss by divine decree, injustice and one named traitor. It isolates the variable you need: the decisive thing was never the artillery, it was the salaried

standing establishment and the fiscal-administrative machine that could reproduce artillery, drill and intelligence at scale — exactly the *compagnies d'ordonnance / taille / Bureau-brothers* artillery train / negotiated-capitulation combination that faced the English in 1449–50. And it gives you an unforgettable image: thirty wagons built in Cairo that changed nothing.

2. **France after Crécy and Poitiers, 1346–57.** Indispensable because it is the same war and the same two states with the roles reversed, so it converts your Normandy chapter from a single observation into a symmetry: in 1346–56 the French explained an English *systemic* victory by rain, Genoese treachery, noble cowardice and sin; in 1449–50 the English explained a French *systemic* victory by treason, the surrender of Maine, bad commanders and unpaid garrisons. Montebelluna's line — that the vices of the feudal army "ne viennent pas de sa structure, mais de la lâcheté avant tout" ([Lefèvre](#)) — is the epigraph the chapter wants, because it is a contemporary explicitly considering and rejecting the structural explanation. And Venette's Genoese passage shows a mechanical cause being transmuted into a treason narrative *within minutes, on the battlefield* (p. 43).
3. **France 1940, used carefully as the modern control.** Value: it shows the pattern surviving into an era of professional staffs and mass literacy, and Pétain's "trop peu d'enfants, trop peu d'armes, trop peu d'alliés" is the purest specimen of input-accounting substituting for system-accounting ([text](#)). Risk: Bloch proves the correct answer was available in real time, so 1940 is contaminated by motive. Frame it honestly as the case where blindness became *motivated*, and use Bloch's "le triomphe des Allemands fut, essentiellement, une victoire intellectuelle" (p. 56) as the standard against which the fifteenth-century observers should *not* be judged too harshly.

Keep the Qing Opium War case in reserve as a sidebar on component-level imitation, and use Prussia 1806, France 1871 and Russia 1906 as the control group in a short methodological note: they demonstrate that the misdiagnosis is not inevitable, and that what breaks it is a permanent professional institution chartered to explain defeat. The English crown in 1450 had no such institution. It had a parliament, a set of impeachments, and the murder of the duke of Suffolk.

7. Evidence status and limitations

- The Normandy 1449–50 baseline used here rests on Anne Curry, "Guns and Goddams: was there a Military Revolution in Lancastrian Normandy 1415–50?", pp. 172–188. Key supporting findings: "by 1449–50 it was the French who [carried the torch] ... with their *francs archers* and their use of field as well as siege artillery in the expulsion of the English" (p. 173); "more commonly in the Reconquest, the French preparations for bombardment and assault, rather than actual bombardment, were enough to prompt surrender" (p. 187); "each place was picked off, one by one, but the fall of the major places led to a domino effect" (p. 187); "many surrenders were forced on the English by the local populations" (p. 187); the English "had been taken by surprise" (p. 186); from the truce of Tours the English had "reduc[ed] the garrison establishment to no more than 2,500 men, thereby putting perhaps 2,000 out of employment," prompting Curry's pointed question, "can it be a coincidence that Charles was able to create his *francs archers* at the very point that the English were releasing large numbers of troops from their service?" (p. 186); and "behind

the English success in earlier decades, and the French success in 1449–50, was political will and also resources" (p. 187). Note that Curry's own conclusion is deflationary about an artillery revolution — "the evidence from Lancastrian Normandy is not supportive of an artillery revolution on the part of the English" and "it is doubtful whether an artillery revolution could have helped the English in 1449" (p. 186) — which strengthens rather than weakens the book's thesis: the operative French advantage was the *system* (money, standing troops, siege apparatus, and above all the credible threat that produced capitulation without bombardment), not the guns.

- I did **not** fetch the Searle and Zhang article from the Taylor & Francis page during this session; the quotations at pp. 89–103 were obtained from a full-text copy of the same article. Before publication, re-verify those page references against the Taylor & Francis version of record.
- The Frieser material is quoted from a review of *The Blitzkrieg Legend*, not from Frieser's book; verify against the monograph before citing.
- Rank 6 (1894–95) is a research lead only.
- Master Roger and Thomas of Spalato page references follow the CEU Press bilingual edition and the *Florilegium* 4 article respectively, as fetched.